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[ycfounders] Fraud

Daniel Sugarman <dsugarman@zentail.com> Thu, Sep 11, 2014 at 7:07 AM
Reply-To: dsugarman@hdtradeservices.com
To: ycfounders@googlegroups.com

Thank you everyone who followed up: David Cann, Tom Sparks, Doug Feigelson, Zaid Farooqui, Benjamin Bryant, Russell Smith, Charlie Pinto, Daniel Conrad, Simon Tisminezky, Kai Stinchcombe, Jesse Cooke, Eric Migicovsky (& James Pierson @ pebble). Thanks especially to Jason Tan and Patrick Collison for speaking to me on the phone so I can better understand your services. I wanted to put together a basic guide to our findings, I hope people find this useful. tldr: manually approve any and all orders.

Fraud & Me (as a startup merchant)

The issue of fraud can be quite serious as a startup merchant, especially selling physical products. Your supplier isn't going to eat the cost, neither will the credit card companies which leaves all the liability on you. If you aren't careful, you will not find out about these issues until the customer issues a chargeback with their credit card companies. This means that you may be selling products for 4+ weeks before you find out you have a problem. A company with low margins, like ours, could lose a month of runway a day and wind up out of business by the time the problem is identified.

How to fight fraud

It is important to keep a close eye on your transactions. There is a fine line between providing excellent service to your good customers and preventing bad behaviour. There doesn't seem to be a better solution than to manually approve every single order at first, and automate more and more as you collect data. The key data points you want to look for are:

- Billing Address
- Shipping Address
- Billing Phone Number
- Shipping Phone Number
- IP Address (geolocated)
- Card Type
- Order Size ($ & Quantity)

Most fraud we see is different billing & shipping addresses. The best review process seems to be googling street addresses, checking white page results (or safecaller.com) on the phone numbers and calling the phone numbers. If you aren't sure, you have a decision to make whether you want to side towards customer service or fraud prevention and that might likely be a calculation with respect to the order size, frequency of fraud and your runway. I am told that verifying billing phone numbers is impossible, otherwise it would be simple to verify a phone number and then hold the package until you receive a response from the customer.

"Friendly" Fraud

Defined as a customer who buys a product and later files a chargeback claim. Make sure that your company name appropriately represents you on your customer's credit card statement. The more data you keep track of, the easier it will be to win one of these claims. If you use Stripe, they will actually eat the cost of the chargeback claim even though they are charged regardless. To win a claim, provide the IP address, carrier and tracking info as well as proof of delivery. If you have it, any engagement with support, discovery, registration and shopping cart should be included.

Sift Science

You should definitely use Sift Science, but it is important to really understand their product, they likely won't protect you very much in the beginning. We use their magento plugin and originally wrongfully assumed that they would automatically prevent orders going through that were fraudulent, this is NOT the case. They provide a machine learning algorithm that will learn your specific attackers patterns as well as more global fraud trends. Their magento plugin will never prevent an order. The Sift Science global model (data collected from all of their users) will likely not be sufficient in catching your fraud attempts. This means that you must see a number of attacks before the model learns what attributes are likely to identify a fraudulent attack. If you have repeated attacks of a certain type, the system will become hypersensitive to your attackers. It will also become hypersensitive to vulnerabilities specific to your platform and customers. **It is very important that you manually label fraudulent orders on your dashboard or the system will never learn.**

Stripe

You all know and love Stripe, they lead the space in ease of connectivity for developers. Stripe is also great for chargebacks because they will pay the fee if you win a claim (normally you are charged either way). Stripe might lead you to a catch 22, do you store credit card info or not. With this info, you are strongly liable. WIthout it, it will be very hard to troubleshoot to a certain degree; it will be downright impossible (without custom work*) to contact the issuing bank to find out more information or to provide the FBI with the information they request. Stripe does not currently provide phone support, do not expect to get enough information from their email support to make a decision on intercepting a package. The issue is that there really isn't much they can do on a complex attack and the time window is so small, just intercept the package. If you can, **hold off on shipping any suspicious order.** I would email patrick@stripe.com right away if you have any major problems, he is very responsive and deeply wants to help YC companies (he even offered custom work if that was the diagnosis).

*It is possible to use Stripe's API to get the issuing bank information. You can be proactive and record this information on every order and contact the bank where necessary. I am told these inquiries don't generally go far.

Looking Into (help if you can!)

- Insurance
- Fraudster Databases
- Auto-reporting with ic3.gov

Our Attack

We have seen a diverse set of attacks from friendly to really technically advanced. Some were easy to detect, ex. IP from foreign address and shipping to a freight forwarder. Our biggest attack was really hard to detect. Our attackers used stolen credit card information to send packages to residential addresses close to the credit card holder. They would also use a proxy to access our website close to the shipping and billing addresses. They would list the billing and shipping phone to be the same phone number and these numbers would be from the correct area as our "customer". Without verification on the billing phone #, we didn't know if these #s are real or fake. Thankfully all of these phone numbers went to the same 2 people.

Select Raw Responses

Tom Sparks - Sounds like you could use IPViking, its a live list of attackers/fraudsters. If you've got a list of the IP's from the folks who've ripped you off, I'll run it against the list and see how many of them would've been caught.

Charlie Pinto - This is what happens when you deal with payments. The best course of action would be to store the cards yourselves, save as much contextual data as you can (geo-location, information, card type, amount, etc.) then ship card data off to stripe.js, and build an in-house fraud system based on the card and contextual data you store. It's really the only way to stop all their fraud bullshit.

Daniel Conrad - My experience selling a physical product (a smart phone) we had to do manual review of transactions, sometimes rejecting transactions after manual review 24+ hours later. Review might involve checking google street view on addresses and other manual activities. Not sure what Stripe/Sift will let you do (I was at Google and using Checkout so had total access) but I don't think there's any way of getting around human input; fraudsters will find the weakness in an automated system by trial and error and unchecked fraud quickly goes to infinity. Law enforcement can't move fast enough. Physical products are different!

Kai Stinchcombe - Nobody has yet built a neural network one one-billionth as powerful as the human brain. When paypal was losing $10m a month to fraud they moved back to human review and fixed the problem. Pull out a step in the automation and let someone review it manually, like set up a pending orders dashboard where someone clicks to release each order. Put down the list of goods ordered, shipping address, billing address, phone number, name, email address, and geolocation of the IP address ordered from, that kind of basic stuff, and it'll be pretty obvious who is legit and who isn't. And if it's unclear, call the billing phone number, the customer will be happy to hear from you either way. The reason you're being targeted is because you're still shipping goods to thieves – if you fix the problem entirely for even just six weeks they'll realize they're wasting their time and go steal from someone else.

On Tue, Sep 9, 2014 at 5:02 PM, Daniel Sugarman <dsugarman@zentail.com> wrote:

> background: we sell physical product. we are dealing with a distributed fraudulent attack, likely orders with stolen credit cards. we thought we were set up well to combat fraud as we have an integration with Sift Science, but all of these orders have extremely low sift scores. what other methods do you use to prevent and detect fraud?
>
> to report fraudulent orders with the issuing banks, we need the credit card information, which we don't store. this info is also blocked out on our stripe dashboard. anyone know how to actually get this information or contact stripe? customer service is virtually non-existent, waiting 6 hours + on their only contact method (email).
>
> what steps do you take to report fraudulent attacks? we have been in contact with our local FBI unit as well as the Internet Crime Complaint Center because this is America and we don't negotiate with terrorists

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